

Time-Variable State Spaces for Auditable ESG and Taxonomy Evidence in Operational Information Systems

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Abstract Environmental, social and governance (ESG) reporting combines operational information that changes across systems and time. Floor areas, energy measurements, emission factors, renovation measures, regulatory criteria and evidence documents may refer to different object scopes and validity periods. A plausible report can therefore be misleading when these differences remain hidden. Carrier elements, valid-time anchors and typed value classes form a common address space. Domain state, rule effect and manifestation remain separate through an explicit evaluation order and six conformance criteria. Reporting reference time is distinguished from coordinate-specific evaluation times, exposing non-synchronous compositions. A two-building example keeps data, evidence and rule versions attached to the same addresses, exposes scope-mixed aggregation, and localises four failure classes without turning a reporting view into a second domain state. The contribution is a specification-oriented modelling approach, not a new ESG indicator, taxonomy standard or database engine. It complements established temporal, analytical, provenance and constraint approaches by specifying a joint contract for state reconstruction, scope selection, rule evaluation, evidence and output generation. Conditional determinism propositions, a technology-neutral implementation mapping and an incremental conformance protocol make the contract testable. Empirical effectiveness remains future work.

Keywords Environmental informatics · ESG reporting · European Union taxonomy · sustainability information systems · time-variable data · state-space modelling · auditability · operational information systems

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1 Introduction

Sustainability information is not created only when a report is produced. It emerges continuously from operational, financial, technical and document-oriented processes. In a building portfolio, relevant information includes floor area, energy consumption, emission factors, renovation measures, investment expenditure, evidence documents and regulatory assessments. These items change at different times and are often maintained by different systems. Nevertheless, they must support sustainability reporting, taxonomy-related evidence, investment decisions and portfolio management.

This creates a specific problem for environmental informatics. An indicator is rarely an isolated formula. It is the result of combining an object population, a time reference, typed values, applicable rules and a reporting purpose. Energy consumption per square metre, for example, is meaningful only if numerator and denominator refer to the same object scope, measurement period and rule context. Otherwise, a report may look technically correct while its domain basis is no longer traceable.

The guiding question is:

How can time-variable sustainability information be specified so that domain state, rule effect and evidence view remain separate while becoming jointly checkable?

The paper makes five contributions. First, it defines a common addressed state space with explicit reconstruction and coordinate-specific evaluation time. Second, it specifies a joint contract for state reconstruction, scope selection, versioned rule effect, evidence and manifestation. Third, it formulates six well-formedness obligations, six execution-conformance criteria and two conditional determinism propositions. Fourth, it exercises the contract on a non-trivial two-building example that exposes four failure classes through localisable rejection witnesses. Fifth, it maps the logical responsibilities to technology-neutral implementation options and provides an incremental conformance protocol.

The full-paper investigation is organised around three more specific questions. RQ1 asks which information must be part of an address so that changing ESG data remains reconstructable rather than merely historised. RQ2 asks which ordering and declaration obligations are required to prevent a technically valid query from mixing incompatible carrier populations, units, times or rule versions. RQ3 asks how these obligations can be tested independently of a particular database, rule engine or reporting product. The method is constructive: the paper defines a specification, derives conditional determinism properties from its obligations and exercises them on a deliberately small transfer example. The example is not an effectiveness study; it is an existence and falsification aid for the proposed contract.

The work addresses United Nations Sustainable Development Goals (SDGs) 7, 11 and 13 through its sustainability use case. Methodologically, auditable information infrastructures also provide a link to SDG 9. The term *auditable* denotes information-system traceability: an output can be related to its input state, time references, rule versions and evidence. It does not assert the legal or substantive correctness of a taxonomy classification.

2 Background and Related Work

Environmental informatics studies information and communication technologies for environmental monitoring, modelling and decision support [7]. The European Union (EU) taxonomy and its delegated acts connect economic activities with technical screening criteria, evidence and disclosure obligations [1, 2, 3]. The Corporate Sustainability Reporting Directive (CSRD), the European Sustainability Reporting Standards (ESRS) and subsequent amendments additionally illustrate that reporting scope and requirements can change over time [4, 5, 6]. A sustainability information architecture must therefore preserve not only values but also the rule version and evidence context under which an output was produced.

Several established approaches address parts of this problem. Online analytical processing (OLAP) and data cubes support dimensional aggregation [8]. Temporal databases distinguish time-dependent facts and, in bitemporal variants, valid time from transaction time [9]. Data Vault and Anchor Modeling preserve histories and support non-destructive evolution [10, 14]. The PROV Ontology (PROV-O) and provenance semirings represent derivations [12, 18], while the Shapes Constraint Language (SHACL) validates graph structures and constraints [13]. Environmental information systems and sustainability platforms provide practical reporting, registry and cooperation solutions [15, 16, 17].

The proposed model does not treat these approaches as deficient. Its claim is narrower: it specifies a joint contract across state reconstruction, scope, rule effect and manifestation. OLAP leaves open whether numerator and denominator were reconstructed over the same scope before aggregation. Temporal databases do not by themselves determine which rule and output definition was applied. Data Vault and Anchor Modeling preserve changes and schema evolution but do not prescribe a shared typed contract for reconstruction, rules and reporting scope. Provenance models record derivations, while SHACL validates structures; neither alone determines the domain reconstruction and evaluation sequence proposed here. Existing storage or reporting technologies may implement the contract, but they are not displaced by it.

The distinction is one of responsibility rather than storage shape. A temporal table can preserve every value version and still permit a report to combine an energy numerator for two buildings with a floor-area denominator for one. A provenance graph can explain which rows contributed to that result without declaring the result semantically admissible. A constraint language can reject malformed graph structures without reconstructing the applicable domain state. Conversely, the present model does not define transaction processing, physical data placement, query optimisation or a universal ontology. It requires an implementation to expose a small set of observable obligations at the interfaces between those capabilities.

This positioning also separates the proposal from a semantic layer placed only above a data platform. Definitions, units, aggregation behaviour and reconstruction rules are attached to value classes in the addressed state model. A report may select and manifest those classes, but it may not silently redefine them. The approach can therefore be implemented through metadata-driven views, generated queries, rule services or

materialised state projections. Its novelty claim concerns the joint contract and its conformance conditions, not a particular execution mechanism.

The approach is therefore not a competing database product. A relational warehouse, lakehouse, event store or knowledge graph may all serve as technical components. Conformance depends on whether the resulting implementation reproduces the declared state, scope, rules and manifestation, not on the product category used.

Methodologically, the artifact-oriented procedure is informed by design-science principles [19]. Recurrent problems in operational reporting architectures were decomposed into state-bearing elements, time references, rule dependencies and outputs. The resulting specification is evaluated here by internal consistency and a constructed transfer example. Independent empirical validation remains future work.

3 Core Concepts and State-Space Perspective

The formal vocabulary is deliberately compact:

- A carrier element is the persistent object at the chosen working granularity, such as a building or technical asset.
- A valid-time anchor is a domain-relevant point in time from which a changed state contribution applies, such as a meter reading or renovation date.
- A value class is a typed semantic role with a declared value domain, unit rule and typed undefined marker. For each carrier on which the value class is instantiated, a reconstruction rule is declared. It is more specific than an untyped table column; examples are annual final energy in kWh, floor area in m² and evidence record status.
- A rule specification is a versioned deterministic derivation, constraint or check with declared inputs and dependencies.
- An evidence record is a versioned reference assigned to a carrier and time context, such as a meter document or energy performance certificate.
- A provenance record *prov* is a versioned relation over state contributions, corrections, evidence records and rule results. A manifested result carries a separate derivation reference to the exact input provenance used to generate it.
- A manifestation is a deterministic, side-effect-free output definition. It reads state and rule results but does not redefine them.

3.1 Addressed Contributions and Reconstruction

Unless stated otherwise, a contribution applies from its anchor up to, but not including, the next anchor; a contribution at the final anchor applies through the remaining valid-time domain. A contribution may additionally carry a declared measurement or reference interval; this describes the observation represented by the contribution and does not introduce a second state-time axis. Here *valid-time anchor* denotes a time point, not the stable entity-key construct called an anchor in Anchor Modeling [14].

Let O be the finite set of carrier elements at one documented working granularity, T a finite ordered valid-time domain, and G the set of typed value classes. Their common address space is

$$D = O \times T \times G.$$

In plain terms, each contribution answers three questions: *Which object? From which domain-valid time? In which semantic role?* The term *tensor* is used only for this multidimensional addressability. It does not imply numerical tensor decomposition [11].

For a carrier o , let $S_o \subseteq T$ contain the anchors at which one of its state contributions changes. To make reconstruction total, the initial anchor $\min T$ belongs to every S_o ; a value class without an applicable initial value carries its typed undefined marker $\perp_g$. The latest applicable anchor at time t is

$$\tau(o, t) = \max\{u \in S_o \mid u \leq t\}.$$

Let $\tilde{f}_{o,g}(u)$ be the typed value stored at anchor u , and let $\rho_{g,o}$ be the declared total and deterministic reconstruction rule for value class g on carrier o . The value visible at time t is then

$$f_{o,g}(t) = \rho_{g,o}\left(\tau(o, t), t, \tilde{f}_{o,g}(\tau(o, t))\right).$$

This equation does not require every calendar day to be materialised. It states that a value is reconstructed from the most recent applicable anchor, its stored contribution and a typed reconstruction rule. A last-value class may simply retain the anchor value. A duration-dependent class may use the elapsed interval. A derived value, such as energy intensity, is not hidden in reconstruction; it belongs to the rule layer because it depends on other value classes.

Corrections are represented non-destructively. Instead of replacing an earlier contribution, a correction is recorded separately for the same carrier-time-value address and receives its own version and provenance. The effective state can therefore use the correction while retaining the earlier contribution for traceability. The exact correction algebra is value-class-specific and outside the compact formal core of this paper.

3.2 Well-Formedness Obligations

The address space alone does not make an instance well formed. Six obligations are required before reconstruction is used by a rule or manifestation.

1. Carrier identity is stable within the declared working granularity. A merger, split or reassignment must be represented explicitly rather than by silently reusing an identifier.
2. Every carrier has an initial anchor at $\min T$. Missing initial values are represented by the typed marker $\perp_g$, not by an untyped null or an undocumented default.

3. For each instantiated carrier-value-class pair, the reconstruction rule $\rho_{g,o}$ is total and deterministic over its declared domain, including its treatment of $\perp_g$.
4. Measurement or reference intervals, units, source versions and corrections remain addressable through the contribution or its evidence and provenance records.
5. A derived value is not stored as if it were an independent observation unless its derivation version and inputs remain identifiable.
6. At every instantiated carrier-anchor-value address (o, u, g) used in reconstruction, the declared version and correction policy selects exactly one effective contribution. If it cannot do so, reconstruction is rejected with a typed conformance witness rather than resolved by technical load order.

These obligations distinguish a state contribution from a generic row. They also provide direct negative tests: an implementation is non-conformant if it cannot distinguish missing from zero, if a key is reused for a different carrier, if reconstruction depends on an undeclared query setting, if competing versions are resolved by technical load order, or if a corrected result can no longer be related to the superseded contribution.

Proposition 1 (deterministic visible state). Let O , T and G be finite, let every S_o contain $\min T$, let the declared version and correction policy select exactly one effective contribution at every instantiated carrier-anchor-value address used in reconstruction, and let every instantiated $\rho_{g,o}$ be total and deterministic. Then $f_{o,g}(t)$ is uniquely determined for every $(o, t, g) \in O \times T \times G$ for which the carrier-value-class pair is instantiated.

Proof sketch. Finiteness and the initial-anchor condition make the set $\{u \in S_o \mid u \leq t\}$ non-empty and finite. Its maximum $\tau(o, t)$ therefore exists and is unique. At the resulting carrier-anchor-value address, the version and correction assumption provides exactly one effective stored contribution. Totality and determinism of $\rho_{g,o}$ then yield exactly one visible value, which may be $\perp_g$. The proposition is conditional: it does not prove that an arbitrary source system satisfies the obligations. Failure to select one effective contribution instead yields the typed conformance witness required by obligation 6.

4 Expanded ESG and Taxonomy-Related Example

Consider a portfolio in which Buildings A and B are separate carrier elements at the chosen working granularity. Table 1 shows both carrier coordinates in one visual row for each common anchor and value class. A value is not merely a number in a reporting column: its carrier, semantic type, unit, validity and source role are part of the state address.

The two evidence-related value classes are deliberately distinct: the first records the status of an identified document, whereas the second assesses whether the evidence set required for the taxonomy-related check is complete.

The example deliberately distinguishes *annual final energy in kWh* from *energy intensity in kWh per square metre and year*. Both might appear as ordinary numeric columns in a report, but they have different semantic roles and aggregation behaviour.

Table 1 Time-variable state contributions and derived results for Buildings A and B.

Anchor	Value class	Building A	Building B	Unit/domain	Role/source
t_0	Floor area	6,000	4,000	m ²	Measurements A1/A2
t_0	Final energy/year	660,000	540,000	kWh/a	Meter periods E1/E2
t_0	Energy intensity	110	135	kWh/(m ² ·a)	Rule r_{int}^{v1} per carrier
t_1	Measure	Heating retrofit	Envelope retrofit	controlled term	Plans M1/M2
t_1	Investment	300,000	200,000	EUR	Approved inputs C1/C2
t_1	Evidence record status	available	pending	controlled status	Documents D1/D2
t_2	Final energy (forecast)	450,000	400,000	kWh/a	Scenarios F1/F2
t_2	Intensity (forecast)	75	100	kWh/(m ² ·a)	Forecast and reconstructed floor area per carrier
t_2	Taxonomy evidence completeness	complete	incomplete	controlled status	Complete for A; required evidence missing for B

The unit belongs to the value-class contract; converting a value to another unit does not create an unrelated attribute, but the conversion rule and target unit must be declared.

Scope consistency is equally important. The two carriers deliberately differ in area, energy demand, renovation measure, investment and forecast intensity. Their additive bases nevertheless form one explicitly declared portfolio scope. Correct aggregation first sums the additive bases over the common carrier scope and only then forms the ratio. In particular, the portfolio intensity is not the unweighted arithmetic mean of the carrier intensities:

$$\begin{aligned}
 I_{t_0}^{A \cup B} &= \frac{660\,000 + 540\,000}{6\,000 + 4\,000} = 120 \text{ kWh}/(\text{m}^2 \cdot \text{a}), \\
 I_{t_2}^{\text{aligned}} &= \frac{450\,000 + 400\,000}{6\,000 + 4\,000} = 85 \text{ kWh}/(\text{m}^2 \cdot \text{a}), \\
 \bar{I}_{t_2}^{\text{unweighted}} &= \frac{75 + 100}{2} = 87.5 \text{ kWh}/(\text{m}^2 \cdot \text{a}) \neq I_{t_2}^{\text{aligned}}.
 \end{aligned}$$

If the energy numerator still contains both carriers while the denominator incorrectly contains only Building A, the scope-mixed result becomes $I_{t_2}^{\text{mixed}} = 850\,000/6\,000 \approx 141.67 \text{ kWh}/(\text{m}^2 \cdot \text{a})$. The three different results, 85, 87.5 and 141.67, separate correct ratio aggregation from an unweighted mean and from a carrier-scope error. The model therefore requires a scope check before the derivation. Depending on available evidence, the implementation must either align both bases through a documented allocation, select a common carrier subset or mark the indicator as not derivable. It must not silently combine the values.

Four versioned rules illustrate the separation between state and rule effect:

1. r_{scope}^{v1} checks that energy and area refer to the same carriers and period.
2. r_{int}^{v1} derives intensity only after the scope and unit checks succeed.
3. r_{ev}^{v1} verifies required evidence types, assignments and document versions.
4. r_{trace}^{v1} produces a taxonomy-related evidence trace containing the applicable rule version, result and provenance.

For the sole purpose of illustrating rule-version effects, two synthetic criterion versions compare the portfolio forecast intensity of $85 \text{ kWh}/(\text{m}^2 \cdot \text{a})$ against thresholds of 90 and 80, respectively. These thresholds are not taken from EU taxonomy legislation and do not establish material alignment.

These rules do not claim substantive taxonomy alignment. An actual assessment must use the technical screening criteria for the relevant economic activity and the applicable regulatory version [1, 2, 3]. The example demonstrates how such criteria can be represented and traced, not whether Building A or Building B satisfies them.

The example exposes four classes of failure that are easily hidden in a conventional report. First, a carrier-scope failure occurs when the numerator and denominator refer to different building populations. Second, a temporal failure occurs when a forecast energy value is divided by an area value reconstructed for an undeclared earlier time. Third, a semantic failure occurs when annual energy and intensity are treated as if they shared the same aggregation rule. Fourth, an evidence failure occurs when a numerical threshold is passed although the required document set is incomplete. The example instantiates the scope and semantic failures numerically. The temporal failure is represented structurally because floor area remains unchanged across the shown anchors; the evidence failure is deliberately non-numeric. More generally, scope, temporal and semantic failures can alter the reported number, whereas an evidence failure can leave the number unchanged while altering whether the result may be used for a declared assessment.

For each failure, the specification identifies a rejectable witness rather than merely issuing a generic data-quality warning. A scope witness contains the two carrier sets. A temporal witness contains the evaluation-time lineage for the affected coordinates. A semantic witness contains the value classes, units and aggregation rules. An evidence witness contains the missing or version-incompatible records. This witness-oriented interpretation is important for auditability because it localises the reason for non-conformance without asserting that the underlying business value is legally correct.

5 State, Rule Effect and Manifestation

The three layers answer different questions:

1. Domain state: Which typed contributions are valid for the selected carriers and anchor?
2. Rule effect: Which versioned rules derive, constrain or reject candidate results?
3. Manifestation: Which indicator, evidence relation or report is produced for a declared purpose?

Throughout this section, t denotes the reporting reference time t_{ref} , not necessarily the evaluation time of every emitted coordinate. The active carrier population is selected at t_{ref} ; any different population-selection time must be declared separately and included in the evaluation-time lineage. For a view q , let $G_q \subseteq G$ be the value classes emitted by that view and let $T_q \subseteq T$ be its admitted reporting reference times. The view declares a total evaluation-time map $t_{\text{eval},q}: G_q \times T_q \rightarrow T$. The active state $X_{q,t_{\text{ref}}}^{\text{act}}$ is assembled

coordinatewise from $f_{o,g}(t_{\text{eval},q}(g, t_{\text{ref}}))$ after selecting the scope-consistent carriers. If $t_{\text{eval},q}(g, t_{\text{ref}}) = t_{\text{ref}}$ for every $g \in G_q$, this construction reduces to a synchronous state snapshot; otherwise it is a declared non-synchronous composition.

Let $R_{q,t_{\text{ref}}}$ be the separately declared applicable rule set and $\prec_{q,t_{\text{ref}}}$ an acyclic dependency relation. Rules are evaluated in dependency order. If more than one topological order is admissible, all admissible orders must yield the same result; otherwise the model must declare an additional deterministic ordering rule.

$$Y_{q,t_{\text{ref}}} = \text{Eval}_{(R_{q,t_{\text{ref}}}, \prec_{q,t_{\text{ref}}})} \left(X_{q,t_{\text{ref}}}^{\text{act}}, \mathcal{E}_{q,t_{\text{ref}}}^{\text{act}} \right).$$

Here $\mathcal{E}_{q,t_{\text{ref}}}^{\text{act}}$ contains the version-valid evidence assigned to the active scope under the declared reading. A rule may be implemented as executable code or a declarative expression. A decision that requires human stewardship is represented as a workflow whose recorded outcome becomes versioned evidence for a subsequent rule. In every case, inputs, version, dependencies and result domain must be declared. A legal-text reference alone is provenance, not executable rule semantics.

For a versioned view definition q , manifestation is

$$Z_{q,t_{\text{ref}}} = M_q \left(X_{q,t_{\text{ref}}}^{\text{act}}, Y_{q,t_{\text{ref}}}, \mathcal{E}_{q,t_{\text{ref}}}^{\text{act}}, \text{prov}_{q,t_{\text{ref}}} \right).$$

The manifestation may be an ESG indicator, an evidence relation, a management view or a machine-readable interface. It is deterministic and side-effect-free: it reads the declared input bundle, does not mutate the domain state and emits its own version and provenance reference. Additive bases are aggregated over the common scope before a ratio is formed. Changing a report layout must therefore not change the underlying indicator semantics.

Proposition 2 (conditional determinism of a manifested result). Assume that Proposition 1 holds for the selected coordinates; carrier selection and evaluation-time mapping are declared and deterministic; the declared reading and the evidence-validity and version policy determine $\mathcal{E}_{q,t_{\text{ref}}}^{\text{act}}$ uniquely; the applicable rule dependency relation is finite and acyclic; every rule is deterministic; the provenance record is obtained through a total deterministic derivation from the selected contributions, corrections, evidence records and rule results; and M_q is deterministic and side-effect-free. If all admissible topological rule orders yield the same rule result, then $X_{q,t_{\text{ref}}}^{\text{act}}, \mathcal{E}_{q,t_{\text{ref}}}^{\text{act}}, Y_{q,t_{\text{ref}}}$ and $\text{prov}_{q,t_{\text{ref}}}$, and hence $Z_{q,t_{\text{ref}}}$, are uniquely determined by the declared view and rule definitions, the carrier-, evaluation-time- and evidence-selection policies, the stored contributions and the stored evidence records.

Proof sketch. Proposition 1 uniquely determines every selected state coordinate. Deterministic carrier selection and evaluation-time mapping uniquely determine $X_{q,t_{\text{ref}}}^{\text{act}}$. The declared reading and the evidence-validity and version policy uniquely select $\mathcal{E}_{q,t_{\text{ref}}}^{\text{act}}$ from the stored evidence records for the active scope. A finite acyclic dependency relation has at least one topological order. The order-invariance assumption makes the resulting $Y_{q,t_{\text{ref}}}$ independent of which admissible order is chosen. The declared provenance derivation then determines $\text{prov}_{q,t_{\text{ref}}}$ from the already selected contributions, corrections, evidence records and rule results. Applying deterministic M_q to these four determined inputs yields one manifested result. If order invariance does not hold, a

Table 2 Retrospective interpretations of a manifested result.

Reading	Declared basis	Meaning of the result
<i>As reported</i>	Original manifestation, input, evidence, rule versions and evaluation-time lineage	Historical output as issued; in synthetic v1, $85 \leq 90$ passes the criterion
<i>Restated</i>	Corrected inputs or evidence under an explicit restatement policy and declared rule version	Revised historical output
<i>Recomputed under current criteria</i>	State at t_{ref} evaluated with a later, explicitly identified rule version	Current analytical assessment; in synthetic v2, $85 > 80$ fails the criterion and is not automatically a restatement

further deterministic tie-breaking order becomes part of the specification; the result must not depend on an accidental execution plan.

This proposition states a reproducibility condition, not material correctness. A deterministic rule can encode an obsolete threshold, and a complete provenance chain can reproduce a substantively wrong classification. The model therefore separates technical conformance from legal, scientific and domain validation.

Table 2 distinguishes three retrospective readings that must remain separate:

Directive (EU) 2026/470 of 24 February 2026 illustrates why these readings must remain separate. It changes the scope of CSRD and Corporate Sustainability Due Diligence Directive (CSDDD) obligations. As a CSRD transition measure, it allows Member States, subject to transposition, to exempt first-wave undertakings or issuers initially due to report for financial year 2024 that fail at least one of the new scope thresholds (net turnover above EUR 450 million and more than 1,000 employees) for financial years starting between 1 January 2025 and 31 December 2026. It does not amend EU taxonomy technical screening criteria and does not by itself require a taxonomy restatement [6].

The prescribed sequence is:

state reconstruction $\rightarrow$ scope selection $\rightarrow$ rule evaluation $\rightarrow$ manifestation.

This sequence is the central difference between a jointly specified information process and a collection of independently plausible tables.

6 Conformance, Implementation and Rollout

Conformance is evaluated for a traceable execution, not only for its final number. In the current model, an execution is conformant if six conditions hold:

1. Address and scope conformance: all used contributions belong to declared carrier, time and value-class domains; numerator and denominator use a common scope.
2. Rule conformance: rules, parameters, versions and dependencies belong to the declared rule model; undeclared hidden inputs are not used.

Table 3 Technology-neutral implementation responsibilities.

Logical responsibility	Possible technical realisation	Required guarantee
Addressed state and anchors	Relational/temporal store, lake-house table, event store	Stable carrier keys, typed values, validity and version history
Semantic registry	Catalogue, ontology or governed metadata model	Definitions, units, aggregation and reconstruction rules for value classes
Rule effect	Rule engine, query logic, graph rules or services	Explicit inputs, versioned parameters, deterministic dependencies
Evidence and provenance	Document store plus provenance relation	Stable references, assignments, versions and derivation links
Manifestation	Analytical query, interface or report service	Declared scope and evaluation order; no hidden state mutation

3. Order conformance: reconstruction, scope selection, rule evaluation and manifestation are executed in this order; ratios are formed only after their additive bases have been aligned and aggregated.
4. View conformance: the manifestation reads only declared inputs, mutates no state and identifies its own version.
5. Provenance conformance: base contributions, corrections, evidence records, rule results and output derivations remain separately identifiable.
6. Evaluation-time conformance: every emitted coordinate declares the evaluation time at which it was reconstructed. Where these times differ within a row, the row is identified as a non-synchronous composition and carries its evaluation-time lineage.

These criteria can be implemented with different technology stacks. Table 3 gives a technology-neutral reference architecture.

A practical rollout need not replace an existing platform. It can proceed incrementally:

1. Select one high-risk indicator and document its carrier scope, anchors, value classes and evidence.
2. Implement the specification in shadow mode, compare results at known scope changes, execute conformance checks and retain divergences as test cases.
3. Extend the registry and expose further manifestations only after they demonstrably reuse the same reconstructed state.

This rollout treats the model as a semantic and behavioural contract above the storage platform. The model is therefore compatible with existing data-management tools, while making their joint obligations explicit.

6.1 Conformance Evaluation Protocol

A reference implementation can test the contract without prescribing a product stack. Each test starts from a declared input bundle and compares an observed execution with an expected result or rejection witness.

1. Anchor reconstruction test: query immediately before, at and after each anchor and verify the expected latest-applicable contribution, including $\perp_g$ propagation.
2. Scope mutation test: remove one carrier from only one additive base and require scope conformance to fail before ratio formation.
3. Evaluation-time test: assign different evaluation times to two output coordinates and verify that the row is marked non-synchronous and carries both times.
4. Rule-version test: execute the same reconstructed state under two declared criterion versions and preserve both results without overwriting the as-reported manifestation.
5. Order test: enumerate admissible topological orders for a small dependency graph. Equal results establish order invariance for that case; unequal results require a declared additional order.
6. Evidence test: keep the numerical state fixed while removing a required evidence assignment and verify that evidence completeness changes independently of the number.
7. Correction test: introduce a correction at an existing address and verify that the effective value changes while both the previous contribution and the correction remain traceable.
8. View-independence test: produce two layouts or interfaces from the same manifestation definition and verify identical indicator semantics and provenance references.

Passing these tests does not validate performance or regulatory interpretation. It shows that the implementation preserves the observable distinctions on which the specification depends. Performance evaluation must additionally report carrier count, anchor density, number of value classes, rule-graph size, evidence volume and the ratio of reconstructed to materialised state.

7 Scenarios and Trade-offs

The same addresses can support scenarios without mixing forecasts with observed state. Forecasts, assumptions and measures receive explicit context and version labels, allowing a renovation scenario to relate energy reduction, investment, technical risk, subsidy conditions and evidence status to the same carriers and anchors. This makes trade-offs between objectives traceable but does not solve an optimisation problem. The paper claims structural readiness for scenario comparison, not optimal renovation planning or predictive accuracy.

This separation is relevant when scenarios are compared with reported history. An observed energy contribution, a forward-looking estimate and a target trajectory may share carrier and value-class semantics while differing in context, evidence and production method. Treating all three as ordinary rows invites accidental aggregation. The state-space contract instead requires the scenario context to be explicit and keeps an as-reported manifestation immutable. A later optimisation component may consume the scenario-specific state and rule results, but its objective function, admissible action space and solution status are additional specifications. No optimisation result follows from the addressing model alone.

The same discipline applies to uncertainty. A point estimate and a probability distribution are different value domains even when they describe the same physical quantity. Confidence intervals, model versions and calibration evidence therefore belong to the value-class or evidence contract. This prevents a management manifestation from presenting a forecast with the visual certainty of an observation while retaining the option to aggregate uncertainty under a separately validated rule.

8 Limitations and Further Research

The current model has six principal limitations.

First, it models domain-valid time. An *as-reported* manifestation can be retained as an immutable historical output, but reconstructing what the organisation knew at a past transaction time, or regenerating that output from a later state, requires a bitemporal extension or immutable versions of all inputs. The evaluation-time map is view-level and carrier-independent; carrier-specific evaluation or population-selection times likewise require an extension. Second, rule dependencies are assumed to be acyclic; cyclic dependencies would require fixed-point semantics. Third, the regulatory catalogue is illustrative and incomplete. Fourth, the example demonstrates internal consistency but not external effectiveness, scalability or user benefit. Fifth, manual stewardship decisions and disputed evidence require an explicit workflow rather than an implicit deterministic rule. Sixth, the present conformance criteria have not yet been implemented as a complete reference test suite.

Further research should combine a reference implementation and machine-checkable conformance tests with empirical cases of controlled scope changes, practitioner interviews, document analysis and a systematic market and literature study. Comparative evaluation should examine whether the specification reduces unexplained indicator differences across reporting views and how much implementation effort it adds. A second research line concerns bitemporal provenance and the governance of rule changes. A third concerns the controlled use of the same state space for scenarios, optimisation and automated evidence retrieval.

9 Conclusion

Time-variable ESG and taxonomy-related evidence requires more than a set of reporting tables. It requires a common contract for the objects being described, the times from which information applies, the semantic type of each value, the rules used in evaluation and the view produced for a particular purpose.

The proposed state-space perspective keeps these concerns separate without disconnecting them. Values and corrections remain attached to stable addresses; rules are versioned and dependency-aware; manifestations remain outputs rather than alternative domain states. Conformance criteria connect this specification to implementable information architectures.

The result is neither a new taxonomy classification nor a replacement for established data platforms. It is a specification-oriented layer that makes their combined treatment of state, scope, rules, evidence and output explicitly checkable. This provides a focused basis for auditable sustainability information systems and for the empirical work still required.

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